

Malawi Nutrition Intelligence System

A Simple Guide for Everyone

What is this system about?

The Malawi Nutrition Intelligence System is a smart computer program that helps the government understand which areas face the biggest nutrition problems and what can be done to solve them.

Think of it like a nutrition doctor for the whole country:

- It looks at data about what people eat
- Finds where the problems are
- Suggests the best solutions

The Main Problem We're Solving

Problem: Government officials need to know:

- Which districts have the worst nutrition problems?
- Should we give vitamin supplements or cash to families?
- Where should we spend limited money first?
- How do we know if our programs are working?

Solution: This system uses artificial intelligence to answer these questions with data instead of guesswork.

How It Works (In Simple Terms)

Step 1: Gather Information

The system looks at real data about all 28 districts in Malawi:

- How much food people eat (calories)
- What vitamins and minerals they get (iron, vitamin A, etc.)
- How vulnerable families are (poverty, education, health)
- Population and location information

Step 2: Find Patterns

The computer learns patterns like:

- Districts with high poverty + low vitamin A = high nutrition risk
- Areas with poor road access have worse nutrition
- Urban areas have different problems than rural areas

Step 3: Make Predictions

For any district, the system can predict:

- Nutrition Risk Score: 0 (very good) to 1 (very bad)
- Risk Level: LOW, MODERATE, or HIGH
- Confidence: How sure we are about the prediction

Step 4: Test Solutions

Before spending money, we can ask:

- What if we give vitamin supplements to 50% of families in Dedza?
- What if we give cash transfers to poor families in Chitipa?

The system predicts how much each solution would help.